George Brooks

Curriculum Vitae

1523 Greene St. #408
Columbia, SC 29225
www.ghbrooks.me

Research Interests

Extremal combinatorics, with interests in spectral graph theory, Turán-type problems in hypergraphs (including the ℓ_2-norm), discrete curvature of graphs, and AI-guided search in large combinatorial spaces.

Education

2020–2025 **Ph.D. Mathematics**, *University of South Carolina*, Columbia, SC
Advisor: Dr. Linyuan Lu

2012–2018 **B.S. Mathematics**, *Saginaw Valley State University*, University Center, MI
Minor in Chemistry

Academic Employment

2025– Present **Postdoctoral Fellow**
NSF Research Training Group in Mathematical Foundations of Data Science
Department of Mathematics, University of South Carolina, Columbia, SC

2025– Present **Adjunct Professor**
Division of Health, Mathematics & Sciences, Columbia College, Columbia, SC

2020–2025 **Graduate Instructor – Teaching Assistant**
Department of Mathematics, University of South Carolina, Columbia, SC

2018–2020 **Adjunct Instructor – Tutoring Center**
Science & Math Division, Northwestern Michigan College, Traverse City, MI

Research Papers

Preprints

G. Brooks and W. Linz. "The Hilton-Milner theorem in the ℓ_2-norm." *arXiv:2610.12433*.

G. Brooks, N. Johnson, W. Linz, X. Liu, L. Lu and W. Vaughn. "On the classification of regular graphs with positive Lin-Lu-Yau curvature." *arXiv:2609.08903*.

S. Agarwal, C. Antley, J. Aulenbacher, G. Brooks, I. Gonzalez, L. Hawranick, W. Linz, L. Lu and A. Williams. "Generalized Nordhaus–Gaddum Inequalities for Eigenvalues." *arXiv:2607.15941*.

Publications

G. Brooks and W. Linz. "Some exact and asymptotic results for hypergraph Turán problems in ℓ_2-norm." *European Journal of Combinatorics* 136 (2026).

G. Brooks, W. Linz, and R. Luo. "Spectral radius and Hamiltonicity of uniform hypergraphs." *Discrete Mathematics* 349.10 (2026).

G. Brooks, W. Linz, and L. Lu. "Maximum spectral gaps of graphs." *Linear Algebra and its Applications* 730 (2026).

G. Brooks, F. Osaye, A. Schenfisch, Z. Wang, and J. Yu. "Outerplanar graphs with positive Lin-Lu-Yau curvature." *Journal of Combinatorics* 16.4 (2025).

G. Brooks, M. Gu, J. Hyatt, W. Linz, and L. Lu. "On the maximum second eigenvalue of outerplanar graphs." *Discrete Mathematics* 348.5 (2025).

Invited Talks

Oct 2026 **LSU Combinatorics Seminar**
Intersecting families in the ℓ_2-norm
Louisiana State University, Baton Rouge, LA

Mar 2025 **AMS Spring Southeastern Sectional Meeting**
Maximum spectral gaps of graphs
Clemson University, Clemson, SC

Nov 2024 **Discrete Math and Combinatorics Seminar**
Maximum spectral gaps of graphs
University of South Carolina, Columbia, SC

Oct 2024 **Graduate Colloquium**
Maximum spectral gaps of graphs
University of South Carolina, Columbia, SC

Feb 2024 **Discrete Math and Combinatorics Seminar**
Outerplanar graphs with positive Lin-Lu-Yau curvature
University of South Carolina, Columbia, SC

Jan 2024 **AMS Special Session on Ricci Curvatures of Graph and Applications to Data Science**
Outerplanar graphs with positive Lin-Lu-Yau curvature
Joint Mathematics Meeting, San Francisco, CA

Nov 2023 **Graduate Colloquium**
On the maximum second eigenvalue of outerplanar graphs
University of South Carolina, Columbia, SC

Mar 2023 **Graduate Mini-Conference in Discrete Mathematics**
Introduction to Flag Algebras
University of South Carolina, Columbia, SC

Oct 2022 **Graduate Colloquium**
Introduction to Flag Algebras
University of South Carolina, Columbia, SC

Workshops

May 2023 **AMS Mathematical Research Communities**
Beaver Hollow Conference Center, Java Center, NY

2022–Present **RTG: Summer School on Mathematical Foundations of Data Science**
Department of Mathematics, University of South Carolina, Columbia, SC

Awards

2023–2024 **Outstanding Graduate Teaching Award**
University of South Carolina – Department of Mathematics

Teaching Experience

University of South Carolina

2021–Present **Instructor of Record**, *Department of Mathematics*
- MATH 344: Applied Linear Algebra (Fall 2026)
- MATH 374: Discrete Structures (Fall 2024, Fall 2025, Spring 2026)
- MATH 141: Calculus I (Spring 2025, Spring 2026)
- MATH 115: Precalculus Mathematics (Fall 2021, Fall 2023–2026)
- MATH 174: Discrete Mathematics for Computer Science (Fall 2022, Spring 2023)
- MATH 344L/544L: Applied Linear Algebra Laboratory (Spring 2022)
- MATH 122: Calculus for BA and Social Sciences (Summer 2021)

2020–2023 **Teaching Assistant**, *Department of Mathematics*
- MATH 142: Calculus II – Honors (Fall 2023)
- MATH 141: Calculus I – Honors (Fall 2021)
- MATH 142: Calculus II (Spring 2021)
- MATH 141: Calculus I (Fall 2020)

Columbia College

2025–Present **Adjunct Professor**, *Division of Health, Mathematics & Sciences*
- MTH 150: Discrete Mathematics (Summer 2026)
- MTH 150: Discrete Mathematics (Summer 2025)

Northwestern Michigan College

2019–2020 **Adjunct Instructor**, *Science & Math Division*
- MTH 111: Intermediate Algebra (Spring 2020)
- MTH 08: Pre-Algebra (Spring 2020)
- MTH 08: Pre-Algebra (Fall 2019)

Service

University of South Carolina

2024–Present **Opportunity Scholars Program – Instructor**
TRIO program supporting first-generation college students.

2024–Present **Partners for Mentoring in Engineering and Computer Science**
Providing gifted high school students academic enrichment in mathematics.

2022–Present **RTG: Summer School on Mathematical Foundations of Data Science**
Teaching and mentoring undergraduate research.

2024–2026 **Math Learning Center Coordinator**
Developing Python labs and facilitating LMS integration for MAP and Gateway testing.

2024–2025 **Discrete Math and Combinatorics Seminar – Organizer**
Inviting speakers and operating the seminar.

2023–2024 **Graduate Student Instructor Peer Mentor Program**
Mentoring graduate student instructors of record.

2022–2024 **Discrete Reading Seminar – Organizer**
Topics have included: spectral graph theory, hypergraph container method, etc.

2021–2022 **Graduate Advisory Council – Student Representative**
Represented graduate students in departmental matters.

Volunteer Work

2023–2026 Lexington-Richland District 5 – Math Outreach

2023–2025 UofSC Annual High School Math Contest – Activity Organizer

Nov 2021 South Carolina Council of Teachers of Mathematics Conference

Oct 2021 Oktoberbest 2021: A Symposium on Teaching